

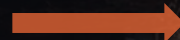
THE INTELLIGENCE LAYER FOR MARS

Harnessed agentic AI for science, engineering, space transportation, communications and settlement

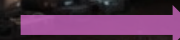
SCIENCE



ENGINEERING



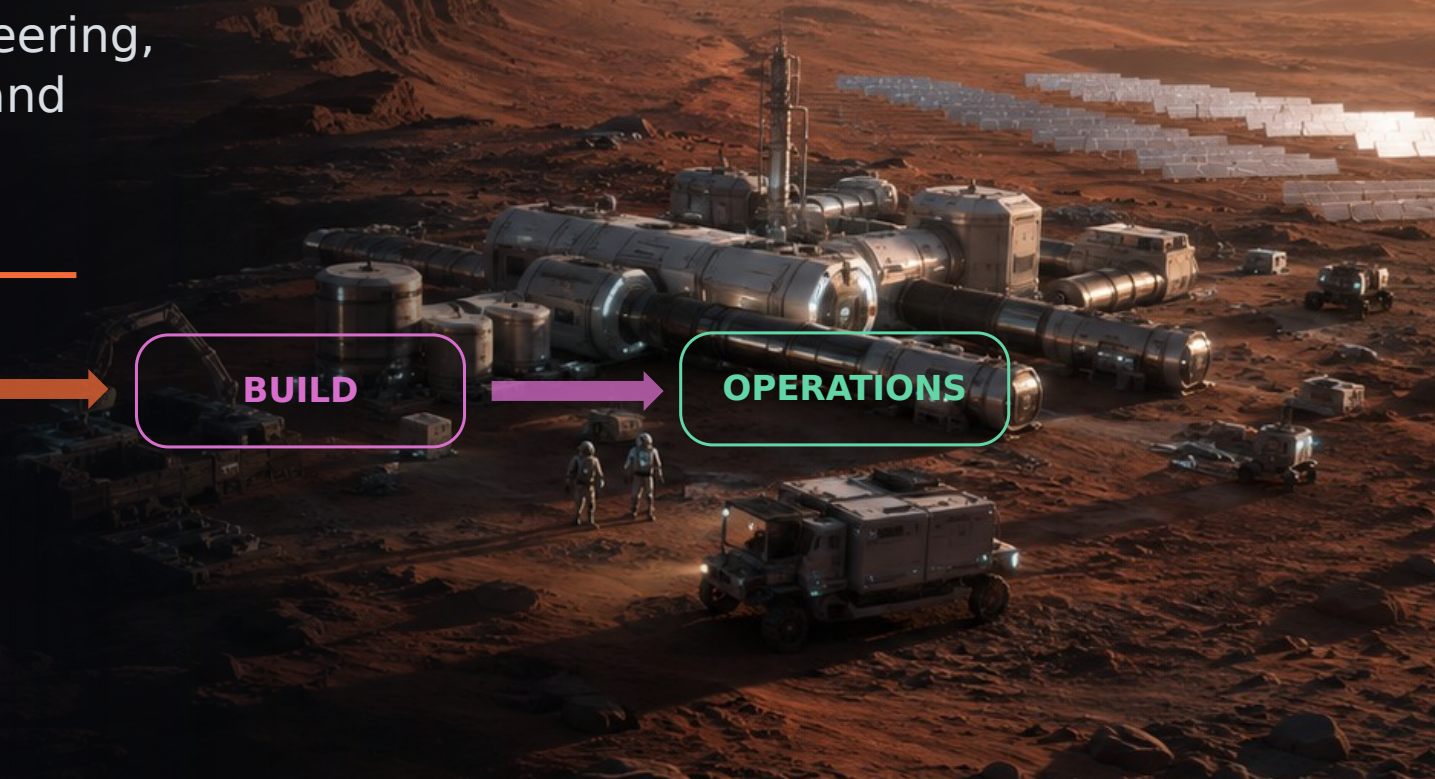
BUILD



OPERATIONS

OCTOBER 22–24, 2026 • USC, LOS ANGELES

BOB J MILLS



Mars is a learning-rate problem

The scarcest resource is time - from a goal to execution.

The critical path is having validated science, using the scientific principles for engineering design and executing the physical implementation of the goal



**Research
And
Development**

**Project
Management**

A Ai agentic harness attacks coordination latency while preserving the evidence needed to close decisions.

Seven functions turn a model into a mission harness

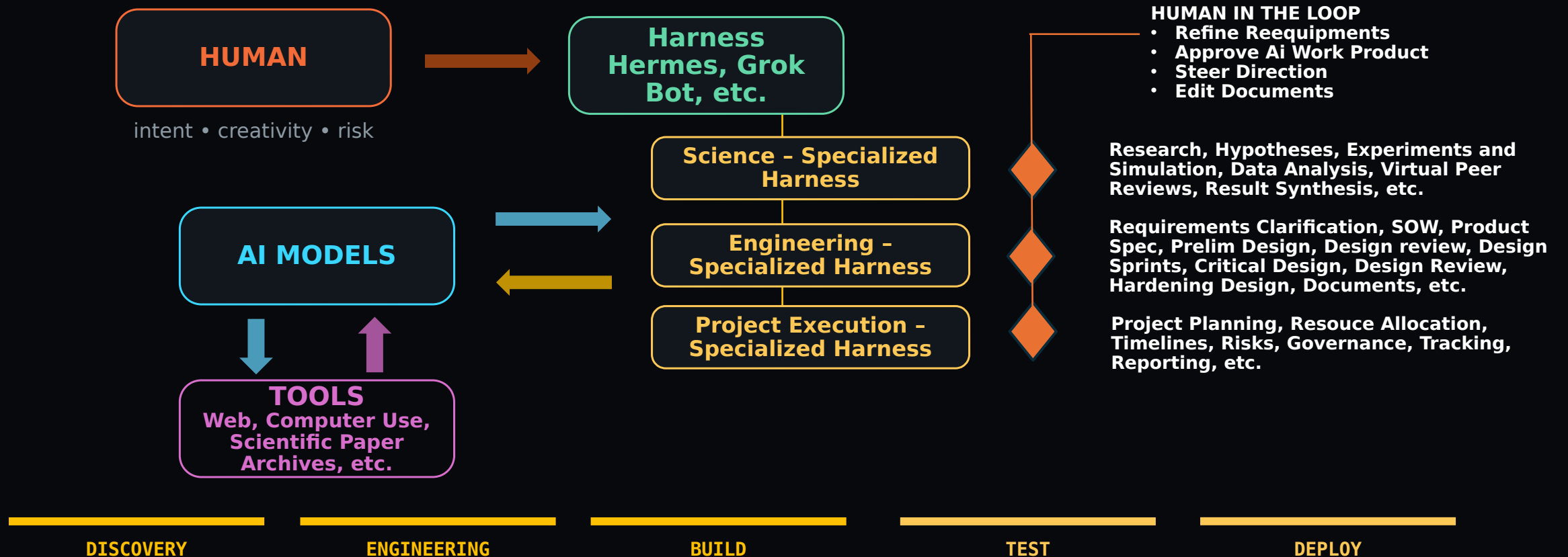
The model supplies reasoning. Seven surrounding functions make its work executable, reviewable and durable.



Stable Skills, Contracts and Consistent Approval authority let every stronger future Ai model inherit a mission-ready way to continue the work.

Skills & Agent identity are durable

Define the capability, evidence and work product first. Then assign the best executor available.



Three harnesses. One digital thread.

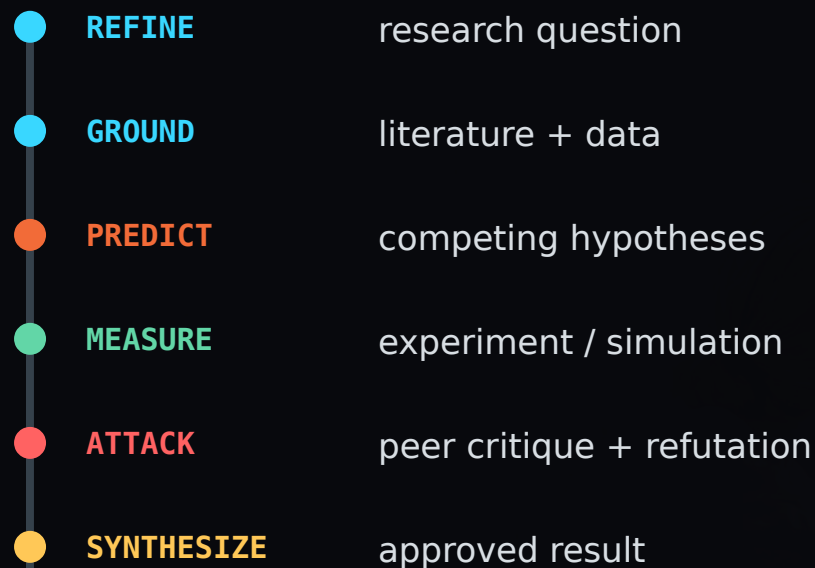
Research, engineering, and execution stay specialized, but evidence and unresolved questions move through one governed baseline.



**Field and test evidence can reopen the science
—or the design.**

Hypatia turns curiosity into defensible science

Falsifiable questions. Explicit variables. Classified evidence. Adversarial attack. Automated Science with Human in the loop

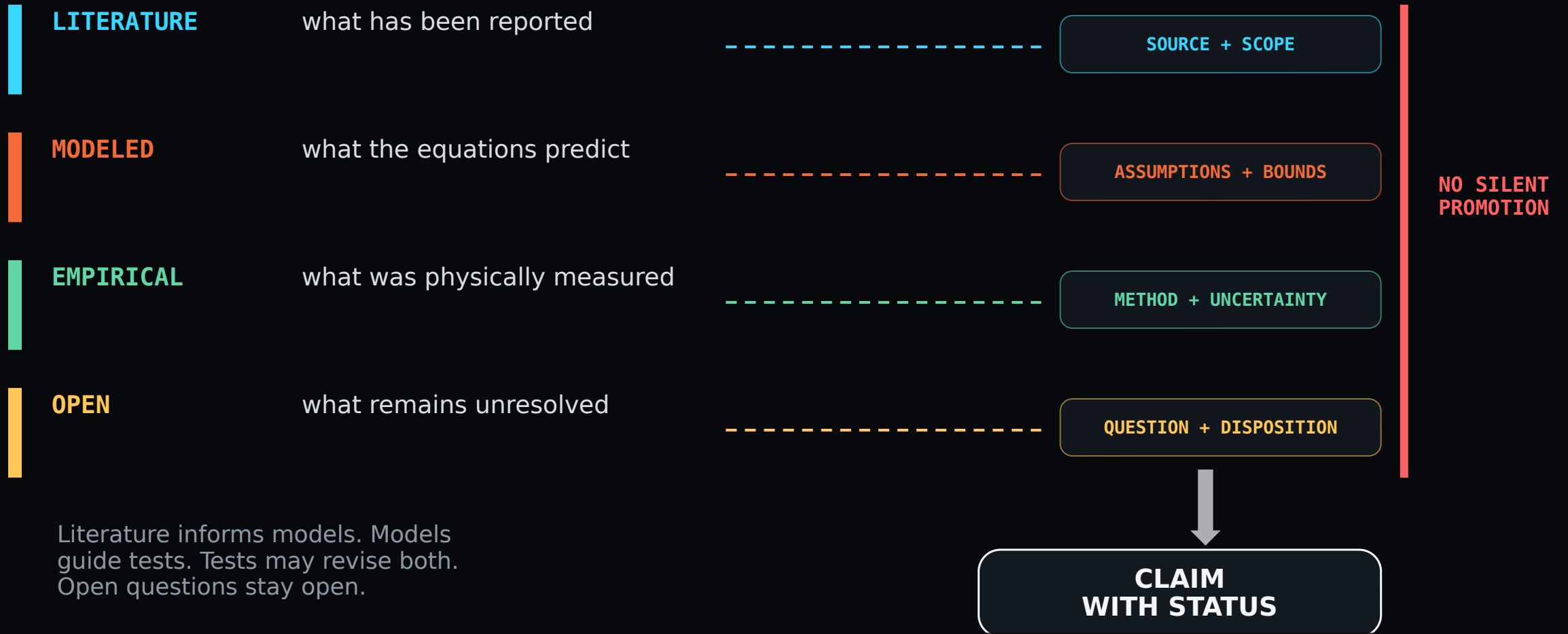


ONLY APPROVED OUTPUTS FLOW FORWARD

GENERATE → VALIDATE →
CRITIQUE
REPAIR → APPROVE →
PERSIST

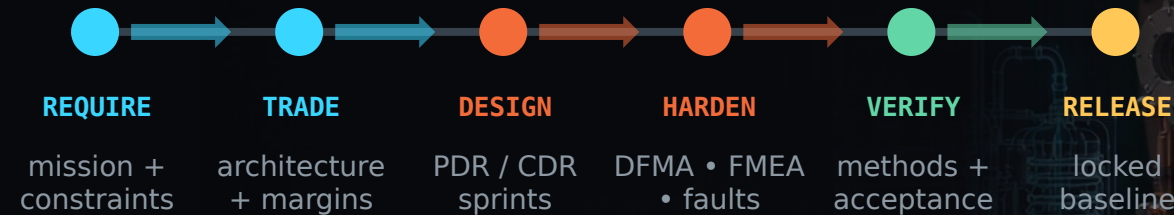
Every claim carries epistemic status

The harness prevents a plausible sentence—or a clean simulation—from silently becoming a physical fact.



Intelligent Engineer turns evidence into a controlled baseline

Drafts accelerate work. Phase gates authorize work.
Automated Engineering with Human in the loop

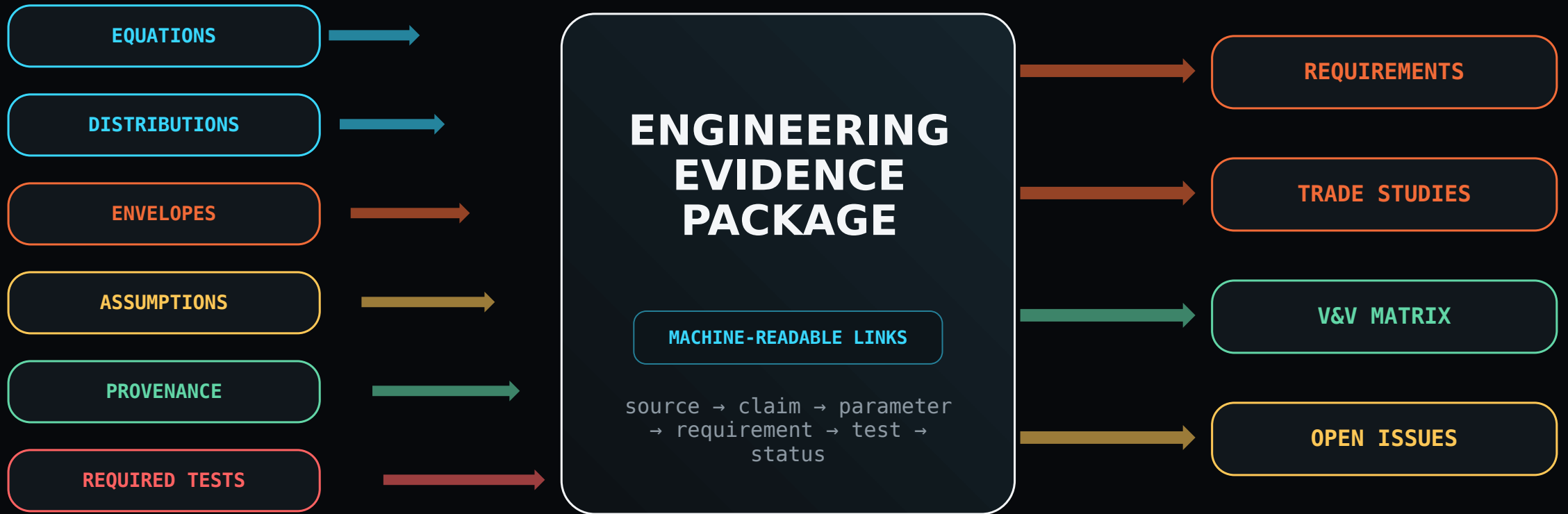


LOCKED BASELINE

Only reviewed phase outputs become authorized inputs downstream.

The handoff is structured—not conversational

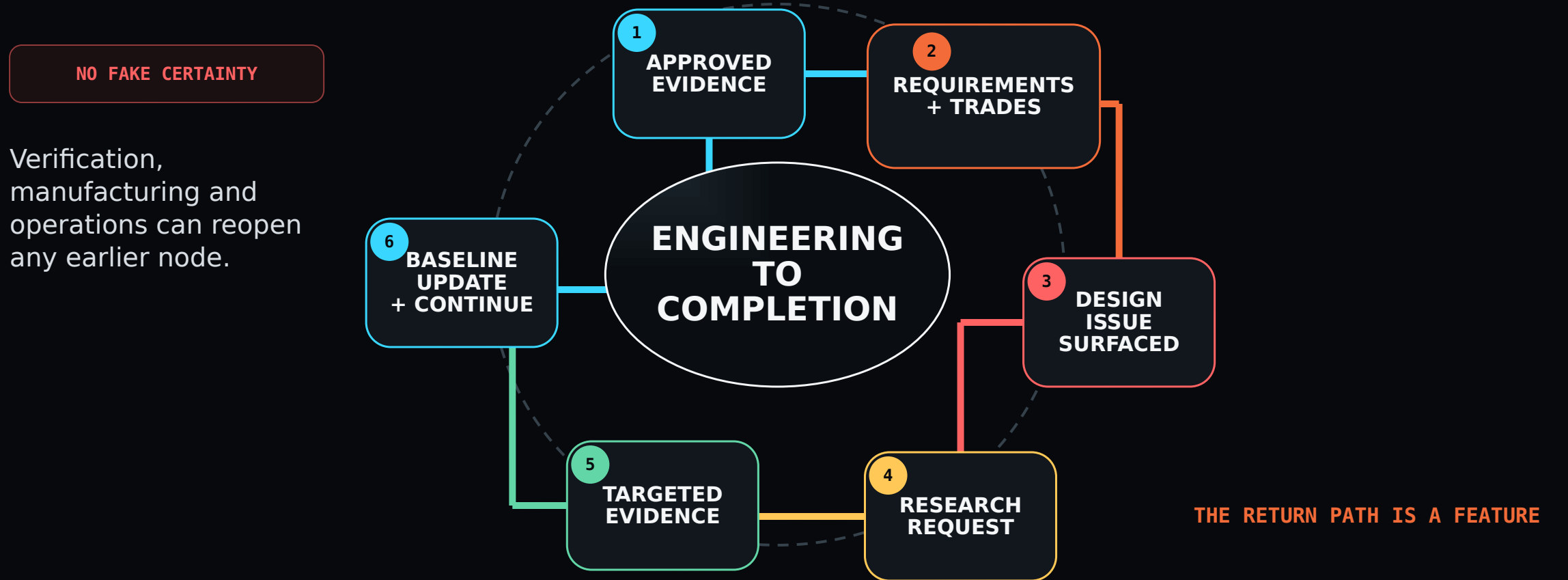
Engineering inherits equations, bounds, provenance and tests together, so changed evidence can reopen the right decisions.



A changed parameter triggers focused rework—not a search through old chat.

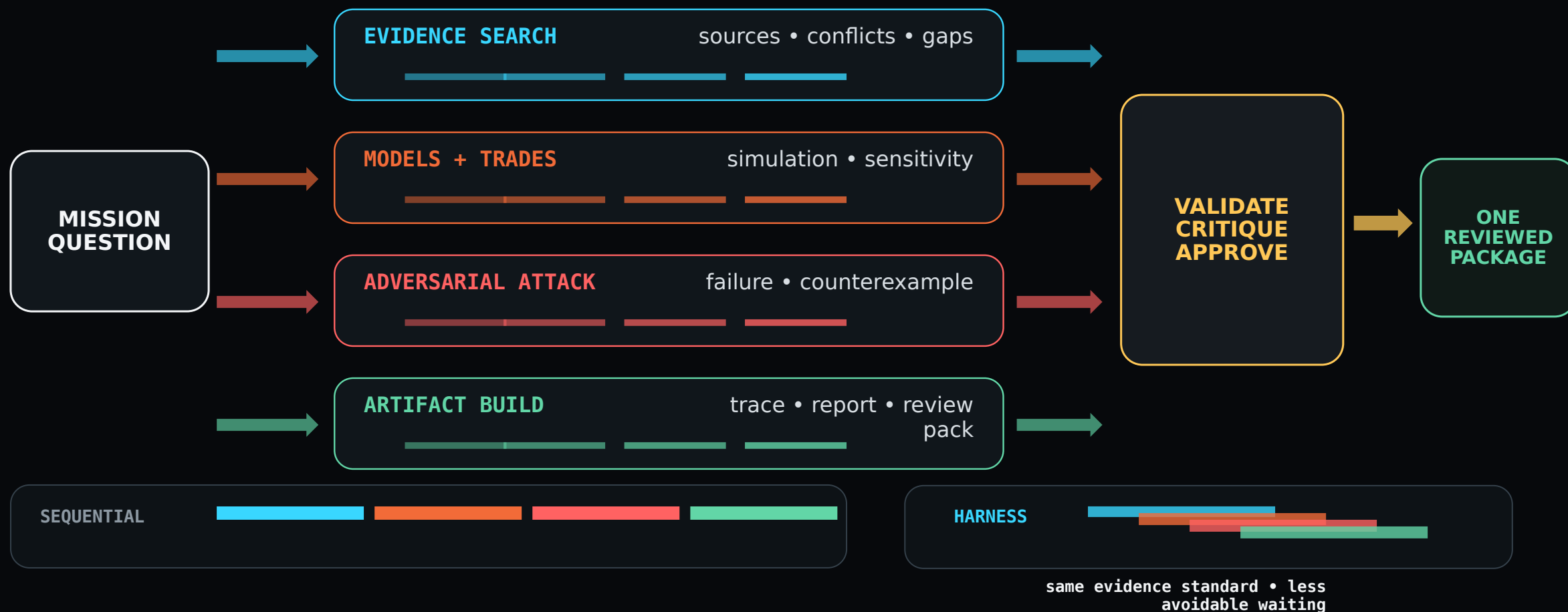
The fastest path to completion includes a return to research

Agents accelerate the issue → research request → evidence return → baseline update cycle instead of hiding uncertainty.



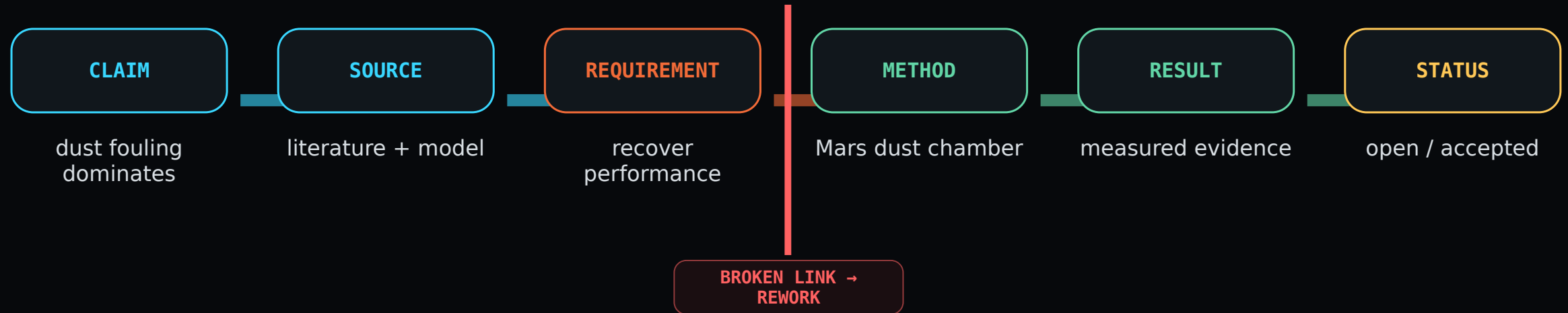
Parallel work shortens waiting; gates preserve truth

The harness coordinates independent searches, analyses and critiques—then merges only reviewed work.



Verification starts at the first claim

Agents maintain the links, expose breaks and launch focused rework before a review gate.



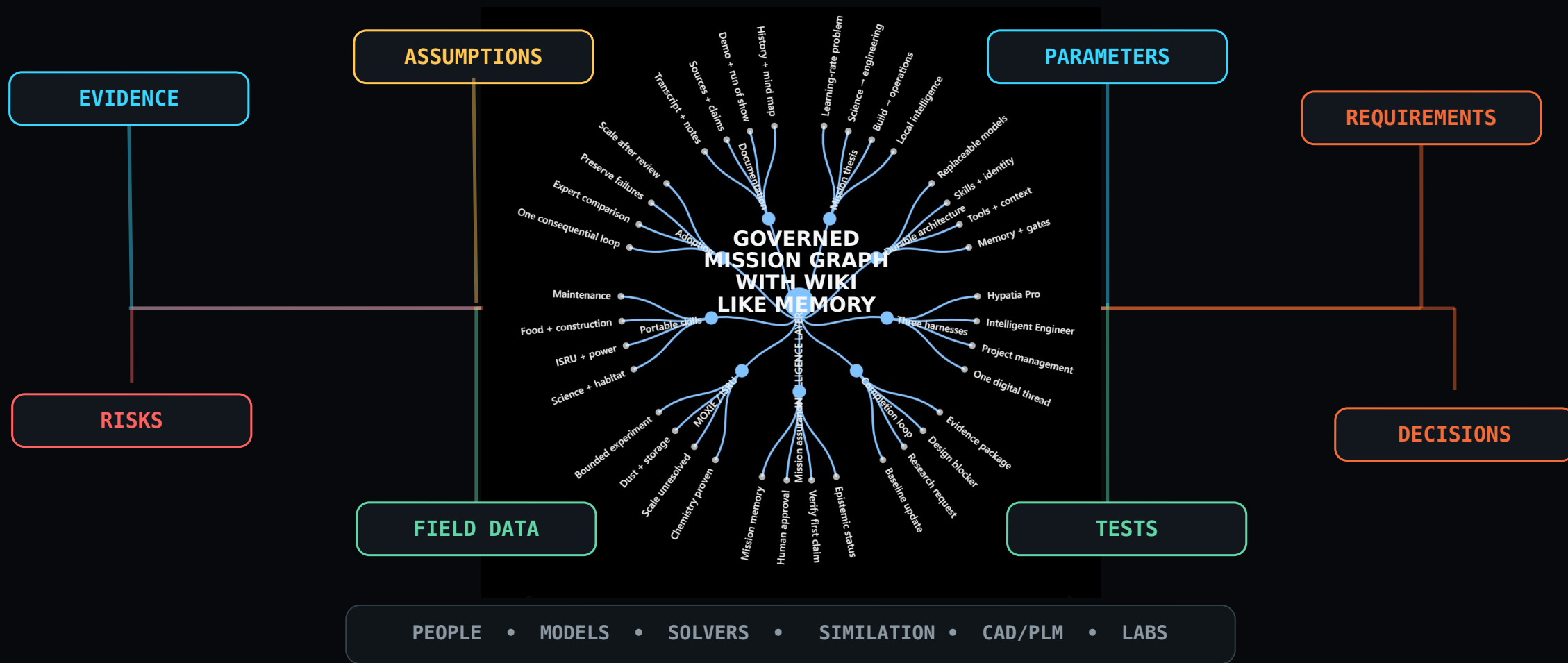
Every uncertainty gets a disposition



AI can assist verification. It cannot make missing physical evidence real.

Mission memory is the compounding asset

A governed knowledge graph preserves what the mission knows, assumes, decides, tests and learns across model generations.



MOXIE proved the chemistry. This modeled scenario tests a settlement-scale oxygen mission.

SCALED MOXIE

304 kWe* 592 t* $\geq 98\%$

Modeled
continuous
power

Modeled O₂ over
730 days

MOXIE
demonstrated
purity

Modeled O₂ allocation, tonnes



Science asks: Is settlement-scale MOXIE physically plausible?

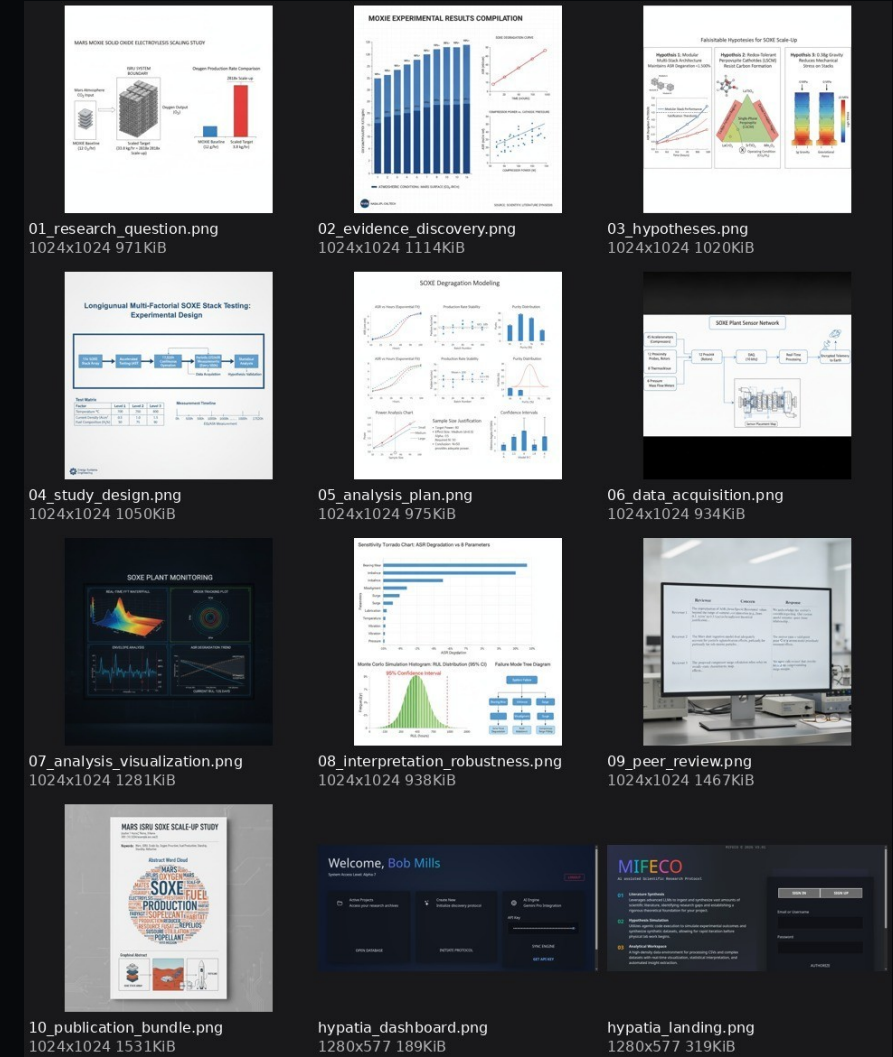
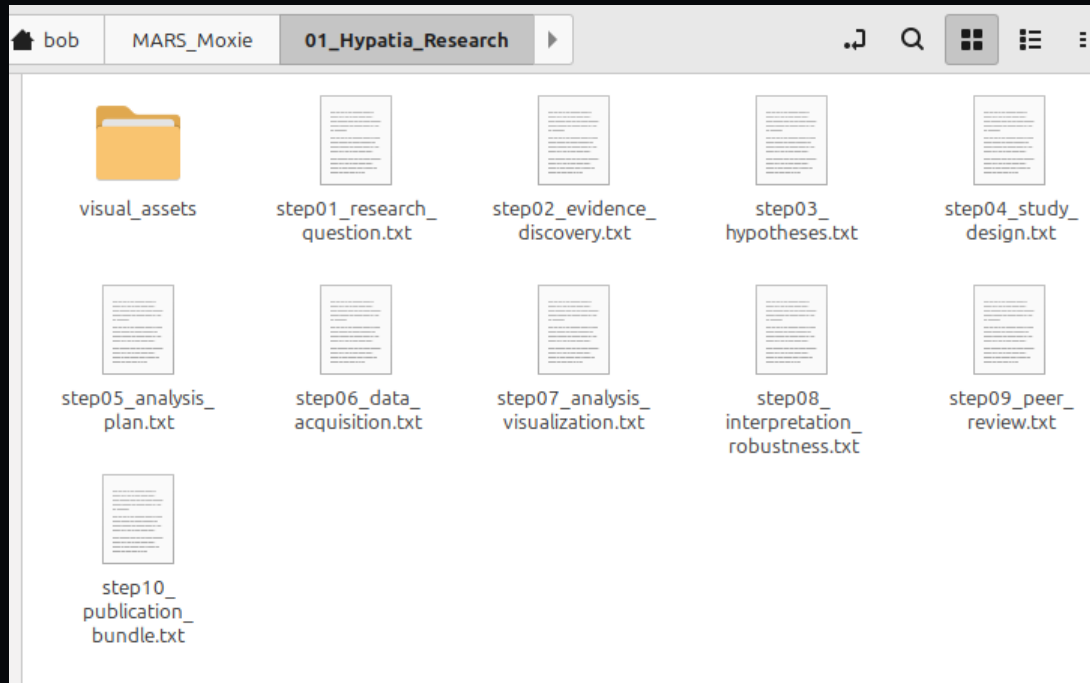
Step 10: Publication Bundle - Project ARES-SOE

****Title:** Industrial-Scale Solid Oxide CO_2 Electrolysis (SOXE) for Human Mars Missions: A 33.8 kg/hr Modular Architecture for Habitat Life Support and Starship Propellant Refueling**

****Author List:** Hypatia AI (Lead Systems Architect), Mars In**

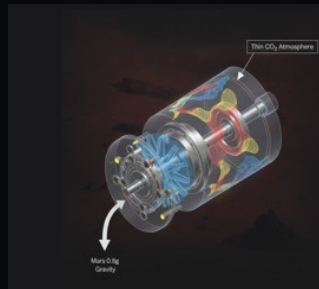
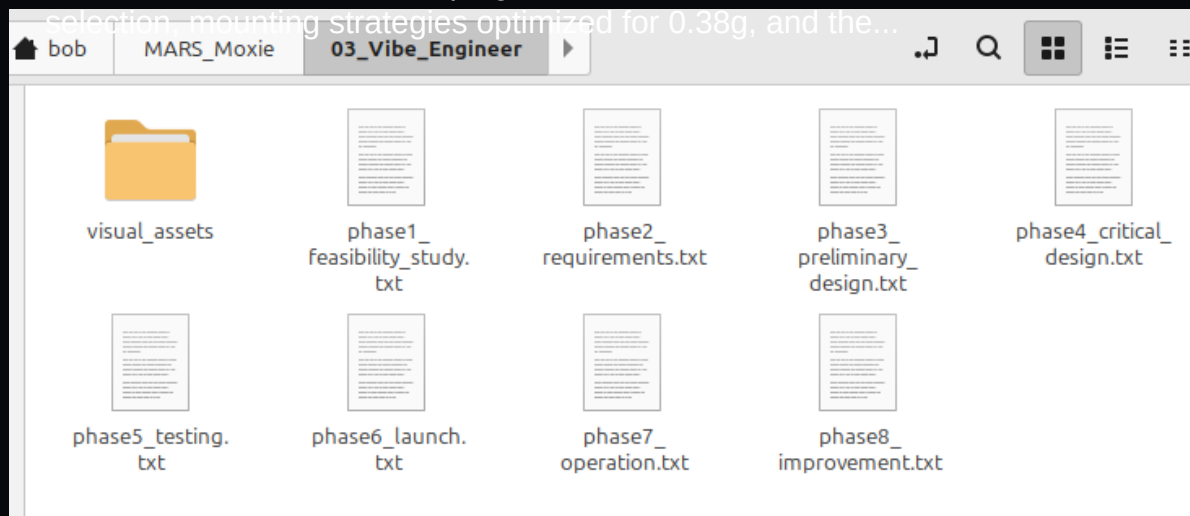
1. Abstract As human Mars exploration transitions from exploration to habitation, In-Situ Resource Utilization (ISRU) must scale by three orders of magnitude. This paper presents the design and validation framework for Project ARES-SOE (Advanced Regenerative Electrolysis System for Sustained

Oxygen Production), a scaled-up solid-oxide CO_2 electrolysis system. Building on the heritage of the NASA MOXIE instrument, the system is designed to produce 592.4 t of O_2 over 2 Earth years (17,520 hours)—sufficient to sustain a 12-person habitat and provide 50% of the...

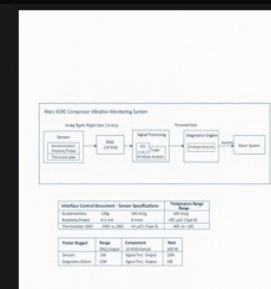


Then the engineering... What would this look like?

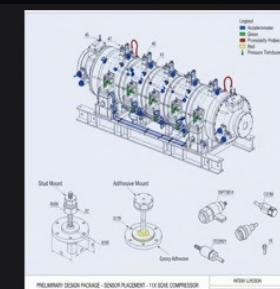
****DOCUMENT ID:**** SOXE-VIB-CDR-004
****PROJECT:**** Mars SOXE (Solid-Oxide Carbon-Dioxide Electrolysis) Scale-Up Plant
****PHASE:**** Critical Design Review (CDR)
****AUTHOR:**** Hypatia | Senior Vibration Lead (VibraEngineer)
****DATE:**** 2024.05.22
****SUBJECT:**** Vibration Monitoring & Structural Integrity Systems for modeled 33.8 kg/h O₂ generation
1. EXECUTIVE OVERVIEW
 The SOXE plant represents an 11x scale-up of the IAC-2022 baseline, necessitating a shift from experimental monitoring to industrial-grade autonomous reliability. With CO₂ feed compressors drawing the bulk of the modeled 304 kWe power budget, vibration is the primary failure mode for the 17,520-hour continuous campaign. This document outlines the final sensor selection, mounting strategies optimized for 0.38g, and the...



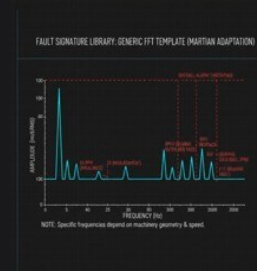
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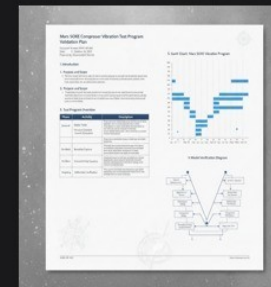
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04_critical_design.png
1024x1024 1454KiB



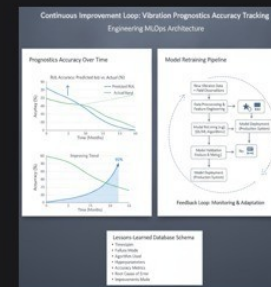
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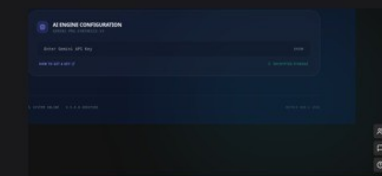
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vibe_engineer_dashboard.png
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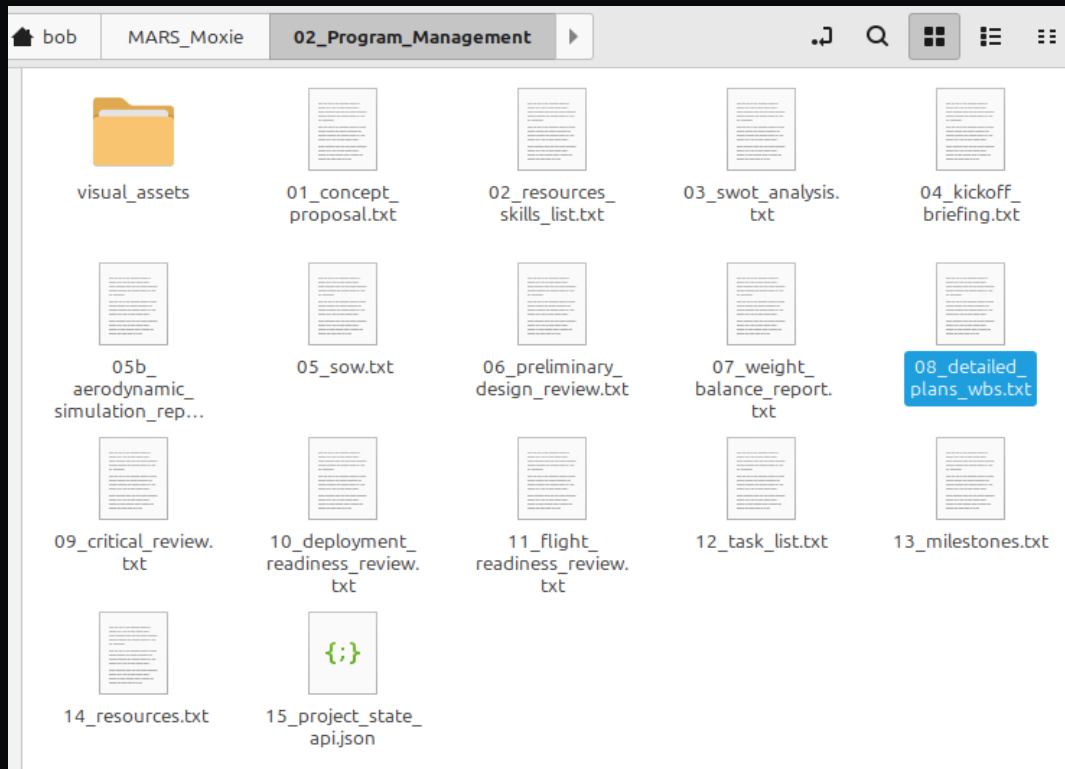
Then a project plan... Resources, Costs, Time, Tracking?

WBS

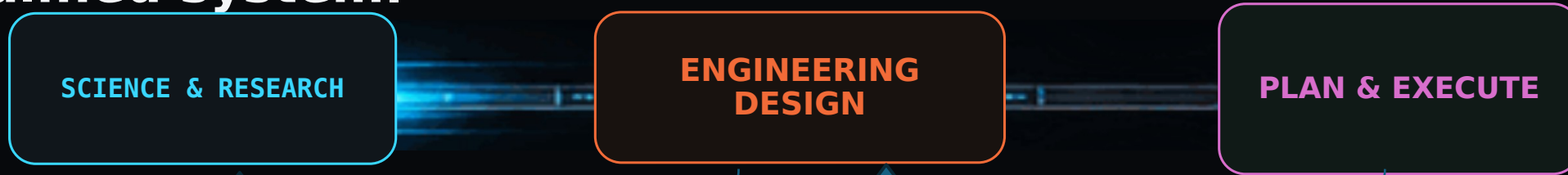
1.0 Atmospheric acquisition + dust rejection
2.0 Solid-oxide CO₂ electrolysis stacks

3.0 Oxygen cleanup, liquefaction + storage
4.0 Modeled 304 kWe power infrastructure

5.0 Integration + mission operations
6.0 Verification + qualification
7.0 Deployment, commissioning + tracking



AI creates an integrated first-pass concept in hours—not a qualified system.

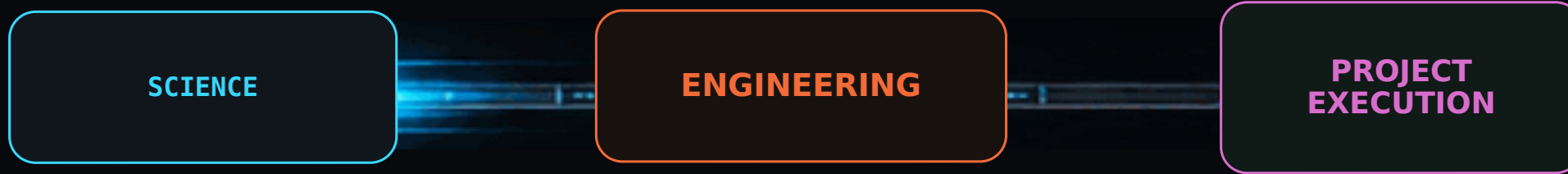


A design or plan-blocking uncertainty becomes an executable Research Request Package—not a vague demand for more study.

- AI interacts between all steps feeding each step context to the next
- AI-driven change management feeds back and re-executes previous steps upgrading documentation
- Every step has its own agentic harness designed to deliver intermediate documents and provide the downstream steps with input
- Every step can run autonomously or can be manually reviewed and edited step by step

HYPOTHESES	REQUIREMENTS	PLAN STEPS
RESEARCH	PRELIM DESIGN	RESOURCES
EXPERIMENT	DESIGN SPRINTS	TIMELINES
ANALYSIS	FINAL DESIGN	PROJECT PLAN
VALIDATION	VALIDATION	TRACK PROGRESS

Five conclusions from science, engineering and project execution



The harnesses compress disciplined concept development. Evidence and authority still determine readiness.

Scale-up is plausible	System integration dominates	WBS connects all systems
Evidence classes must stay explicit	304 kWe remains modeled	Cost + schedule need basis
33.8 kg/h remains modeled	Requirements need traceability	Risks remain visible
Degradation needs experiments	Reliability needs qualification	Changes propagate through state
Open questions drive research	Design must return blockers	Approval gates control release

Workflows become portable mission skills

Codex, Claude Cowork, Hermes and Grok and their future versions can provide the agentic workflows needed captured as Skills and Agent Identities. Below is a small sample of where this could be beneficial.



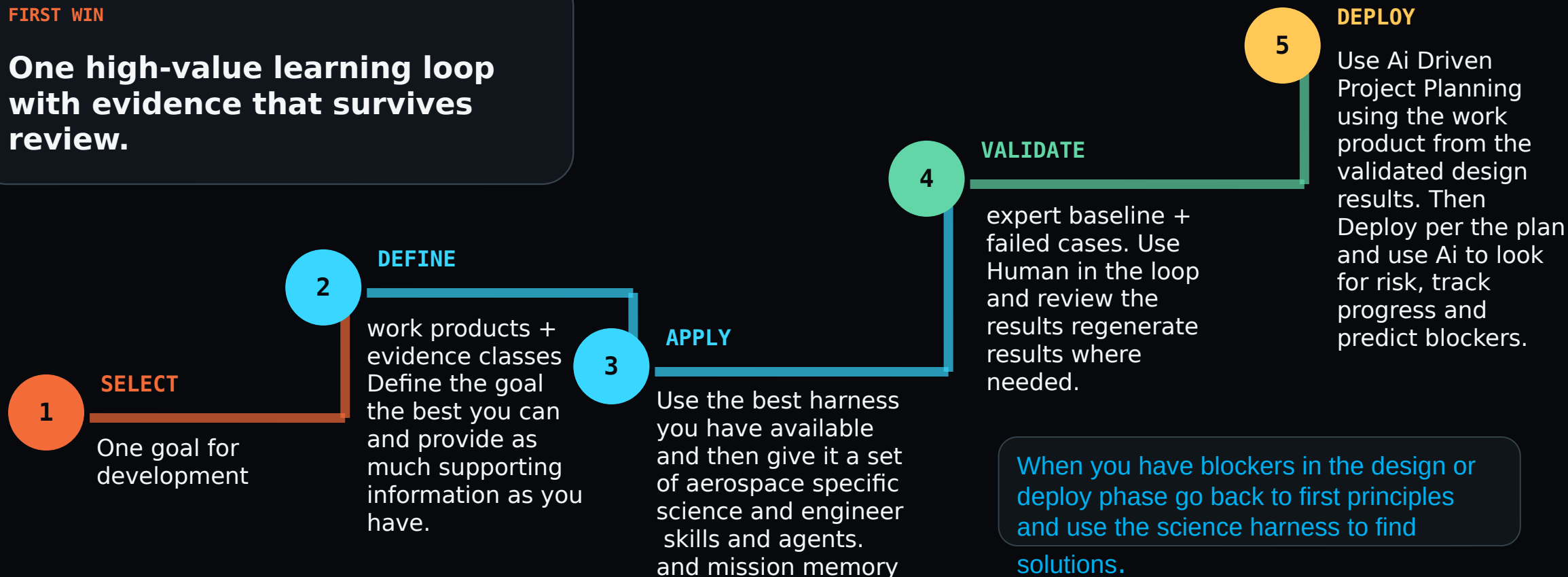
RUNTIMES: CODEX • CLAUDE COWORK • HERMES • GROK

Start with one consequential loop

Prove traceability and expert equivalence on a recurring decision before attempting an enterprise-wide ecosystem.

FIRST WIN

One high-value learning loop with evidence that survives review.



MAKE ENGINEERING A LEARNING SYSTEM

Mars will reward teams that learn faster without losing discipline.



Build the intelligence layer now—so every stronger model inherits a mission-ready way to work.

The Beginning - Not the End

- The speed of Improvement of Ai is exponential and tomorrow it will be able to do much more than today. But the need for specialized skills and agent identities will still exist.
- Even now - if you describe a skill you need, the goal it needs to support and tell an Ai to build a detailed skill it will do it better than any human.
- What I have described exists, works and is tested. The website below will provide free working code as either a set of skills for your choice of harness or as a complete Software application. (You will need to provide your own Ai as this is very token intensive, I do have a paid version to cover the cost of providing a integrated Ai solution.)
- More information and code are available on the <https://www.mifeco.com> website.
- I can be contacted at mifecoinc@gmail.com

Yes, this works on Mars, the Moon, the Space Economy and on Earth but I am most interested in accelerating the way to Mars.

Currently we are on the brink of AGI (Artificial General Intelligence) This is where Ai is more capable than all humans in every subject including knowledge (done) and reasoning (close). We need to work hard to stay in the loop and not be overwhelmed. This is a tsunami that will not slow down or stop. I hope I can help in a small way using Ai to keep humans in the loop while solving the BIG issues to make the world (and out of this world) a better place

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